CentralUniversityof Jammu

Rahya-Suchani(Bagla),District:Samba–181143,Jammu(J&K)

03 March 2025

RollNo. [redacted] B E C S E [redacted]

SEMESTERMIDTERMEXAMINATIONS

Programme	:	**B.Tech. CSE**	**Semester**	:	**VI**
CourseName	:	**Big Data Analytics**	**CourseCode**	:	BECSE3C007
Duration	:	**1.5 Hours**	**Credits**	:	**03**
Maximum Marks	:	**45**	**Minimum Marks**	:	**18**

Afterpursuingthiscourse,thestudentwillbeableto:	
CO01	Recall the fundamental concepts of Big Data and explain its role in addressing business challenges.
CO02	Describe the components of Hadoop and evaluate the functionality of tools in the Hadoop ecosystem.
CO03	Demonstrate the ability to access, process, and manage data using distributed file systems like HDFS.
CO04	Design and develop efficient Big Data solutions by leveraging Hadoop and its associated tools.
CO05	Assess data analysis challenges and implement machine learning models

Part A **VeryShortAnswerTypeLogicalQuestionsof04marksrelatedtodifferentCOs**		**Total Marks 5x2=10**	
S. No.	**Statement**	**Levels**	**CO**
A1.	What is Data Analytics, and how does it differ from traditional data processing?	L2	CO-01
A2.	List and briefly describe the key characteristics of Big Data.	L2	CO-01
A3.	Differentiate between data mining and data visualization in analytics.	L2	CO-05
A4.	Define Big Data and explain its significance in modern computing.	L1	CO-01
A5.	What is the role of Hadoop in data analytics?Why is Hadoop considered a foundational tool in Big Data analytics?	L2	CO-02

PARTB **ShortAnswerTypeAnalyticalQuestionsof6marksrelatedtodifferentCOs** **(Attempt any Five)**		**Total Marks 5x3=15**	
B1.	Describe how data exploration helps in building machine learning models.	L3	CO-05
B2.	How can data visualization techniques improve the interpretation of complex datasets? Provide an example.	L4	CO-05

B3.	Explain the importance of HDFS in processing large datasets.	L2	CO-03
B4.	Why is distributed file storage essential for managing big data?	L2	CO-03
B5.	What are the advantages of using a distributed file system over a traditional file system for big data analytics?	L3	CO-03
B6.	Explain how organizations derive value from Big Data. Provide an example.	L3	CO-01
Part C **LongAnswerTypeDescriptiveQuestionsof20marksrelatedtodifferentCOs** **(OnlyTwoquestionstobe attempted)**		**TotalMarks 2x10=20**	
C1.	Explain the steps involved in data mining for Big Data Analytics. What challenges are commonly faced in this process?	L5	CO-05
C2.	Describe the components of Hadoop and their roles in big data processing.	L3	CO-02
C3.	Discuss the key programming paradigms and tools used in Big Data processing and Big Data Analytics. Discuss in brief about at least two tools used for big data analytics.	L2	CO-02